

## User Stories – Sprint 2

The following user stories define the functionality to be developed during Sprint 2.

### Person Identification

1. As a user, I want the system to detect people in its environment so that the robot knows when someone is present.
2. As a user, I want the system to be able to identify one specific person from a group so that the robot can recognise who it should interact with.
3. As a developer, I want detected people to be shown with bounding boxes so that I can check whether the system is detecting people correctly.
4. As a developer, I want the system to extract body landmark coordinates from detected people so that their movements can be represented accurately.
5. As a user, I want the system to identify the target person using a unique visual identifier so that the robot can recognise them from other people nearby.
6. As a user, I want the system to identify people from a live camera feed so that people can be detected as they appear in the robot's environment.

### Gesture Recognition

1. As a user, I want the system to detect my hand gestures so that the robot can respond to my movements.
2. As a user, I want the system to recognise at least seven predefined gestures so that I can give the robot a variety of instructions using gestures.
3. As a developer, I want recognised gestures to be displayed with their frame number so that I can check whether gestures are being recognised correctly.
4. As a user, I want the system to distinguish between normal hand movements and intended gestures so that the robot only responds to actual commands.

### Motion Mirroring

1. As a developer, I want the system to receive live human motion data so that the user's movements can be processed as they occur.
2. As a developer, I want the detected human joint coordinates to be mapped to the robot's coordinate system so that the human pose can be adapted to the robot's body.
3. As a developer, I want the mapped motion data to be converted into robot joint angles so that the robot can perform the detected human movements.

### Person Following

1. As a user, I want the robot to follow the identified target person so that it can move with them as they move through the environment.
2. As a user, I want the robot to maintain a constant distance from its target person so that it does not move too close or too far away while following.